

Embedded Systems & IoT April -July 2026

Lab Exercise: Schematics with ESP32

Circuit a

Resistors: $R_1 = 20\ \Omega$, $R_2 = 30\ \Omega$, $R_3 = 50\ \Omega$

Supply voltage: $V_s = 125\ \text{V}$

1. Total resistance

$$R_{\text{total}} = R_1 + R_2 + R_3 = 20 + 30 + 50 = 100\ \Omega$$

2. Total current

$$I_{\text{total}} = \frac{V_s}{R_{\text{total}}} = \frac{125}{100} = 1.25\ \text{A}$$

3. Current through each resistor

$$I_{20} = I_{30} = I_{50} = 1.25\ \text{A}$$

4. Voltage drop across each resistor

$$V_{20} = I \times R_1 = 1.25 \times 20 = 25\ \text{V}$$

$$V_{30} = 1.25 \times 30 = 37.5\ \text{V}$$

$$V_{50} = 1.25 \times 50 = 62.5\ \text{V}$$

5. Power dissipated by each resistor

$$P_{20} = V_{20} \times I = 25 \times 1.25 = 31.25\ \text{W}$$

$$P_{30} = 37.5 \times 1.25 = 46.875\ \text{W}$$

$$P_{50} = 62.5 \times 1.25 = 78.125\ \text{W}$$

Circuit b

Resistors: $R_1 = 20\ \Omega$, $R_2 = 100\ \Omega$, $R_3 = 50\ \Omega$

Supply voltage: $V_s = 125\ \text{V}$

1. Total resistance

$$R_{\text{total}} = 20 + 100 + 50 = 170\ \Omega$$

2. Total current

$$I_{\text{total}} = \frac{125}{170} = 0.735294\ \text{A}$$

3. Current through each resistor

$$I_{20} = I_{100} = I_{50} = 0.7353\ \text{A}$$

4. Voltage drops

$$V_{20} = 0.7353 \times 20 = 14.706\ \text{V}$$

$$V_{100} = 0.7353 \times 100 = 73.529\ \text{V}$$

$$V_{50} = 0.7353 \times 50 = 36.765\ \text{V}$$

5. Power dissipation

$$P_{20} = V_{20} \times I = 14.706 \times 0.7353 = 10.8133\ \text{W}$$

$$P_{100} = 73.529 \times 0.7353 = 54.0659\ \text{W}$$

$$P_{50} = 36.765 \times 0.7353 = 27.0333\ \text{W}$$